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Seismic waves
See page 164

1731

Earthshaking discovery

Inventor Nicholas Cirillo used a pendulum to measure earthquakes in Naples, Italy. The amplitude of the pendulum's sways (the extent of their back-and-forth movement) indicated where the earth tremors were most intense. His device was the first seismograph.



Page from Linnaeus's *Systema Naturae*

Stretchy stuff

While in the Amazon rainforest, French explorer Charles de la Condamine came across the substance rubber, which is obtained from a rainforest tree called *Hevea brasiliensis*. He sent samples back to Europe.

Incision made in the tree's bark

Rubber latex fluid collected from tree

1731

Female pioneer

Italian academic Laura Bassi was the first woman to hold a university post in science when she was appointed Professor of Anatomy at the University of Bologna, Italy. A year later, she was also made Professor of Philosophy.



Laura Bassi

1735

Classifying life

Swedish botanist Carl Linnaeus divided the natural world into three kingdoms—animal, plant, and mineral. In his book *Systema Naturae*, he introduced the binomial (two-name) system that classified plants and animals by genus and species. This system is still used today.



1730

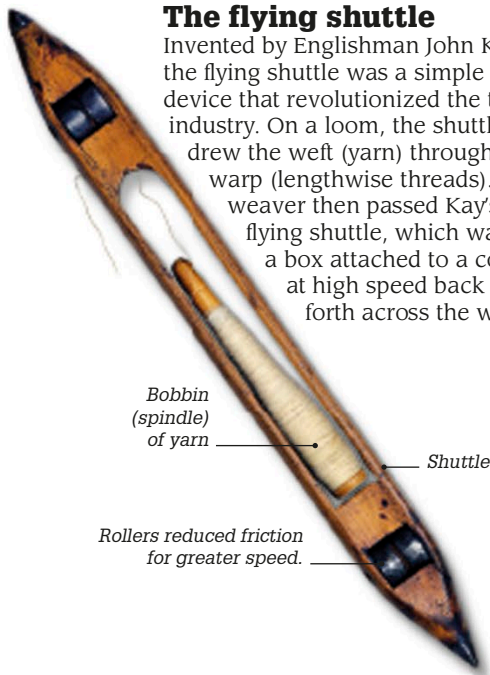
1734

1738

1733

The flying shuttle

Invented by Englishman John Kay, the flying shuttle was a simple device that revolutionized the textile industry. On a loom, the shuttle drew the weft (yarn) through the warp (lengthwise threads). The weaver then passed Kay's flying shuttle, which was in a box attached to a cord, at high speed back and forth across the warp.



Bobbin (spindle) of yarn

Shuttle

Rollers reduced friction for greater speed.



Bornite, an ore containing cobalt

1735

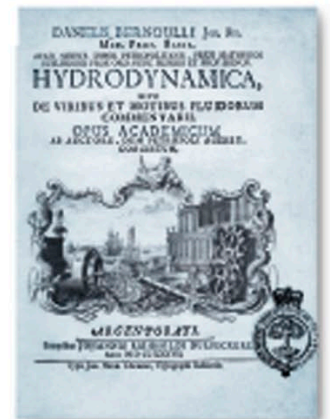
Goblin ore

Georg Brandt, a Swedish mineralogist, identified the element cobalt, which is present in Earth's crust in combination with other minerals. The name cobalt comes from the German word *kobold*, meaning "goblin ore."

1738

Bernoulli's principle

Swiss mathematician Daniel Bernoulli stated that as the speed of a moving fluid (liquid or gas) increases, the pressure within it decreases. His principle explains how an aircraft gains lift because air flows faster over the top of its wings and slower underneath.



Bernoulli published his principle in a book, *Hydrodynamica*



Pitot tubes, invented in 1732 to measure how fast rivers flow, are still used to measure airspeed on aircraft.